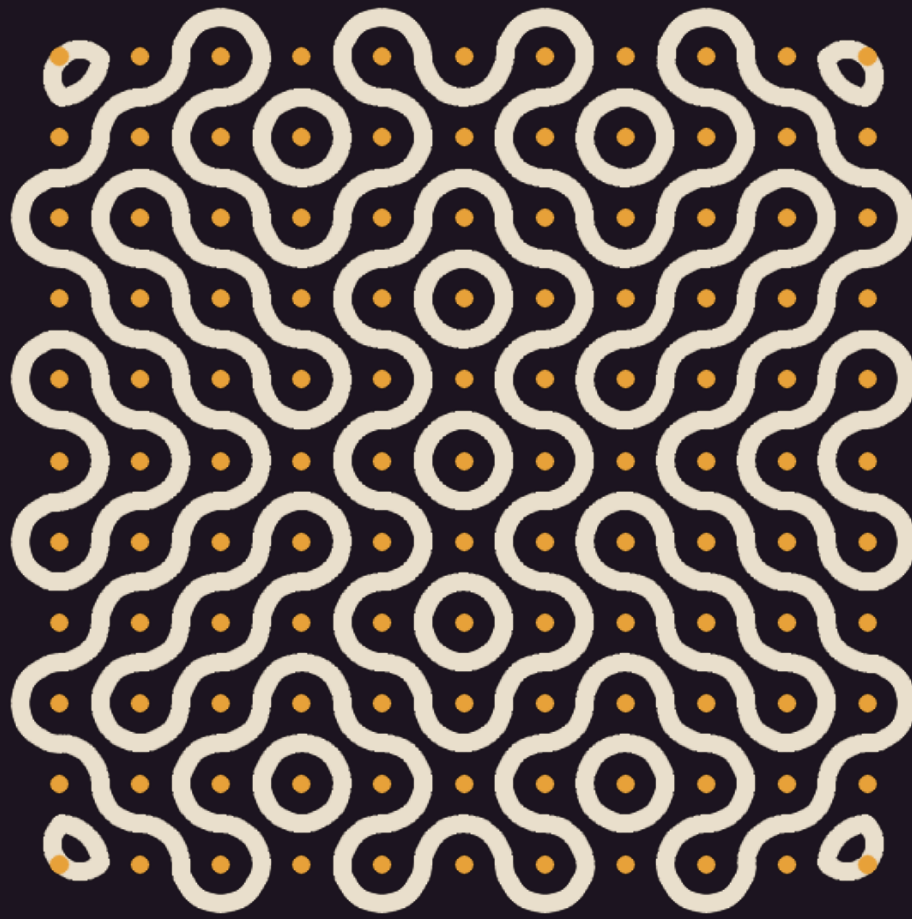
SIH12507 · AICTE · HERITAGE & CULTURE

Kolam Studio

Reading the hidden mathematics of a vanishing art — identify it, recreate it exactly, and generate it anew.



Live: **kolam-studio.onrender.com**

Team Kolam · Smart India Hackathon 2025

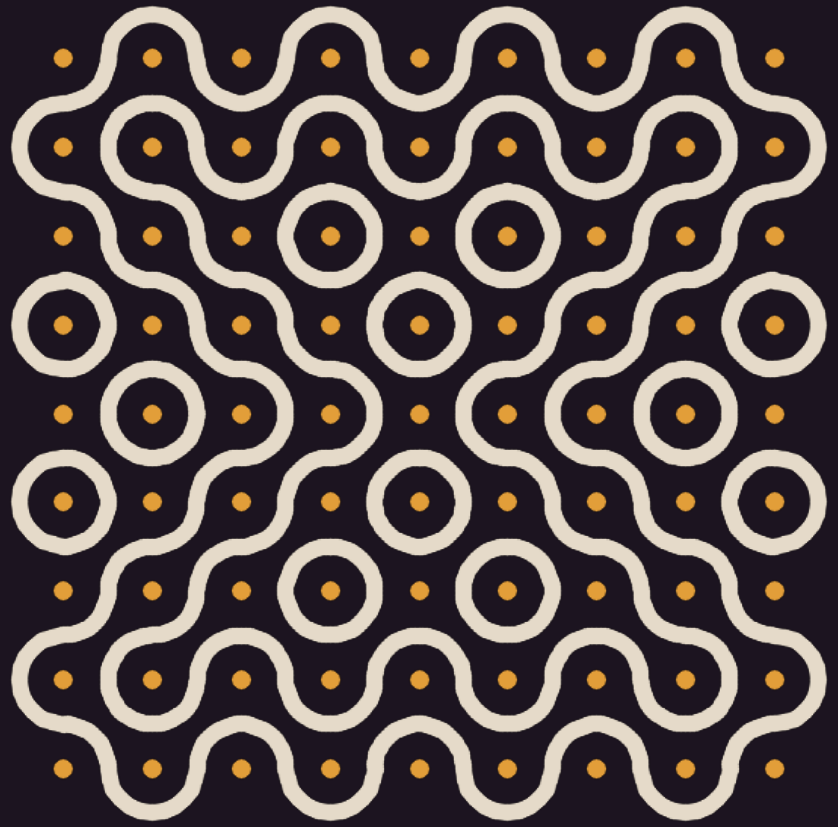
THE PROBLEM

A living art that's quietly disappearing

Across South India, a kolam is drawn fresh at every doorstep at dawn. The art survives in photographs — but the knowledge of HOW each pattern is constructed (its dot grid, its symmetry, its single unbroken loop) is fading with every generation. We can photograph a kolam, yet we have never captured the RULES that make one.

PROBLEM STATEMENT — SIH12507

“Develop computer programs to identify the design principles behind Kolam designs and recreate the kolams.”



UNDERSTANDING THE ART

A kolam is hidden mathematics

Rice flour, a grid of dots, and one looping line — but underneath sits real structure that a computer can read and reproduce:

1. **Pulli**

   A precise lattice of dots — the skeleton of every pattern.

2. **Symmetry**

   Mirror and rotational rules give the design its balance.

3. **Single loop**

   One continuous line that ideally closes without lifting the hand.



OUR SOLUTION

One tool, three moves

1

Identify

Upload or pick a kolam — it is analysed in the browser: dot grid, symmetry and structure are detected, and non-kolam images are rejected.

2

Recreate

The exact kolam is reconstructed cell-by-cell from inferred tile orientations, drawn as closed loops with a glowing pen-head animation.

3

Generate

Endless new, original kolams under the same rules — including a single unbroken line and custom dot layouts.

HOW IT WORKS

From a photo to a principle — and back

Kolam image
upload / sample

›

In-browser analyzer
dots · symmetry · tiles

›

Design principles
structured data

›

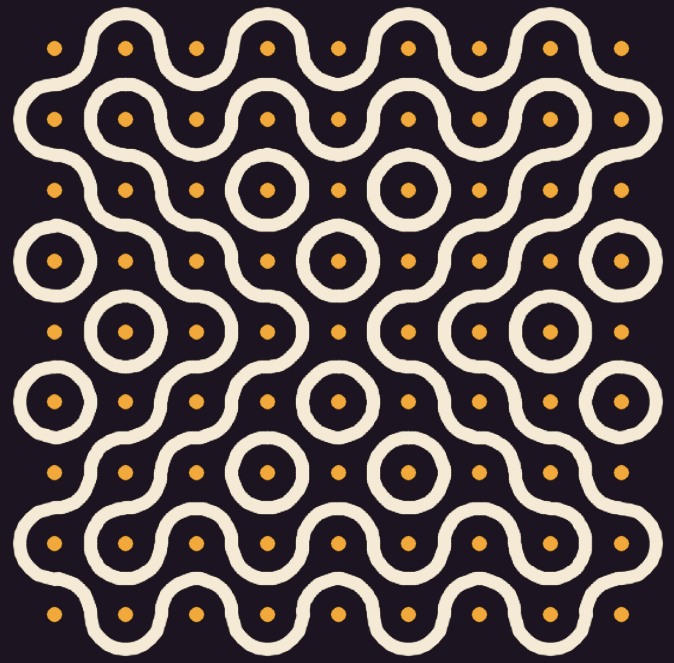
Tile generator
closed-loop canvas

Everything runs client-side — no server, no cold-start.

Python + OpenCV (reference) → **vanilla JavaScript analyzer (shipped, in-browser)**

FEATURE · IDENTIFY & RECREATE

It doesn't copy pixels — it understands



Input

→

Detects dot grid (pulli), cells and symmetry type

Infers the tile orientation of every single cell

Rebuilds the exact kolam, cell-for-cell, as closed loops

Tile-inference accuracy on clean kolams: 100% (verified on the test set)

FEATURE · GENERATE

Because it knows the rules, it can create



Single unbroken line



Custom dot layouts (Square / Diamond / Plus + editor)

TECHNOLOGY & ENGINEERING

Built to be real, fast, and robust

Tile-grammar engine

One curved tile per cell, joined into guaranteed closed loops around the dots.

Single-loop solver

Tile-flip loop-merging reduces the whole kolam to one continuous line.

Fully client-side

Analyzer ported to vanilla JS — no backend, deployable static, no cold-start.

Exact reconstruction

Per-cell tile inference reproduces an uploaded kolam cell-for-cell.

Photo fallback

Illumination-normalise + Otsu + line-period estimate when dots aren't visible.

Offline PWA + SVG

Installable, works offline; export crisp vector kolams. Pytest + CI.

LIVE & DEPLOYED

Not a mock-up — try it right now

- **Runs in any browser**
  desktop or phone, nothing to install
- **Works offline**
  installable PWA, no network needed
- **Open source**
  code on GitHub, with tests + CI
- **No cold-start**
  static hosting — the demo can't time out



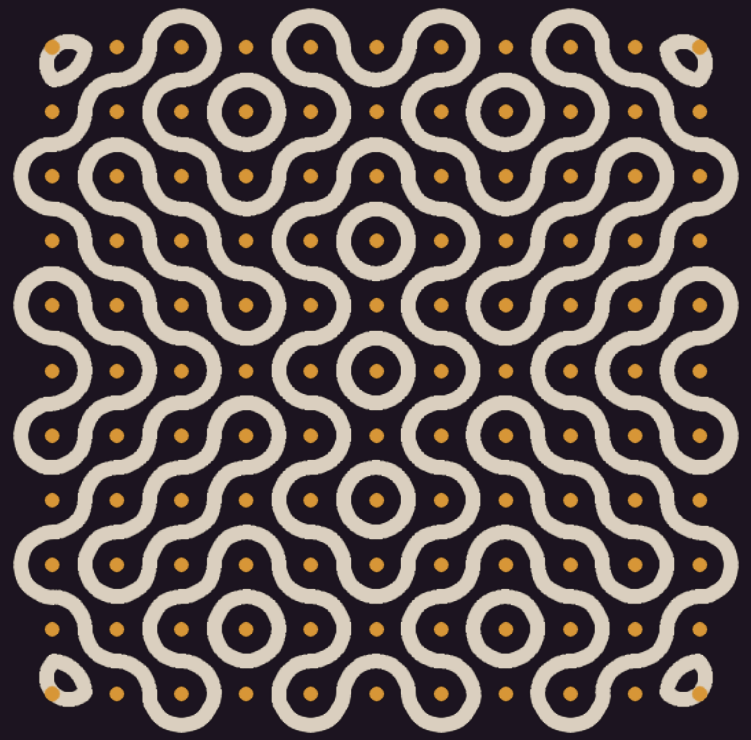
Scan to try

kolam-studio.onrender.com

IMPACT & WHAT'S NEXT

Preserving the rules, not just the pictures

- **Heritage** — Archive the design rules of a fading tradition, not just images.
- **Education** — Teach symmetry, grids and topology through a familiar cultural lens.
- **Design & craft** — Generate motifs for textiles, rangoli, packaging and decor.



Roadmap: sharper real-photograph handling (RANSAC grid-fit, perspective) · regional kolam styles · learn a motif from one example.

"We didn't just preserve a heritage — we taught a machine to keep drawing it."